

Nikhil Singh

CONTACT INFORMATION	Room 515, Building G 26 Campus Kaiserslautern	<i>Contact:</i> niksingh@mpi-sws.org, +4916095870710 <i>GitHub:</i> https://github.com/ciniks117
CURRENT POSITION	Postdoctoral Researcher Max Planck Institute for Software Systems, Germany Supervisors : Prof. Sadegh Soudjani and Prof. Rupak Majumdar Project : AutoCyPher https://www.auto-cypher.eu/ September 2025 – Present	
EDUCATION	Indian Institute of Technology, Kanpur, PhD, Computer Science and Engineering, Jan 2020 - Sept 2024 (9.50/10) Thesis : Synthesis of Safe and Efficient Feedback Controllers for Complex Dynamical Systems. Advisor: Prof. Indranil Saha Indian Institute of Technology, Kanpur, MS, Computer Science and Engineering, Jan 2018 - Dec 2019 (9.0/10) Thesis : Specification guided Automated Debugging of CPS Models. Advisor: Prof. Indranil Saha National Institute of Technology, Jamshedpur, B.Tech, Computer Science and Engineering, Aug 2010 - May 2014 (8.70/10) Project: Intelligent Web Search Engine.	
RESEARCH INTERESTS	Reinforcement Learning Robotics and Automation Formal Methods and Software Engineering Control Theory	
EXPERIENCE	FARE Fellow, IIT Kanpur Safe Reinforcement Learning based Controller Synthesis. Jan'25 - Aug'25 Visiting Researcher Max Planck Institute for Software Ssystems, Germany Nov'24 - Jan'24 <i>Host: Prof. Rupak Majumdar</i> Safe Learning using Kernel based Embeddings. Senior Student Research Associate FMSAFE Project, IIT Kanpur <i>Sponsor: MHRD IMPRINT</i> Jan'18 - Dec'19 Developed an Automated Tool for Debugging CPS Models. Software Engineer OptumSoft, Bangalore June'14 - Jun'16 Software Development for Distributed Systems.	
PUBLICATIONS	* Rupak Majumdar, Nikhil Singh , Sadegh Soudjani. “ <i>Kernel-Based Safe Exploration in Deep Reinforcement Learning.</i> ” Accepted to the Learning for Dynamics & Control Conference (L4DC 2026), University of Southern California, (USC), Los Angeles, USA, June 17-19, 2026. * Rupak Majumdar, Mahmoud Salamati, Nikhil Singh , Sadegh Soudjani. “ <i>Learning Metastable Dynamics.</i> ” Accepted to the ACM International Conference on Hybrid Systems: Computation	

and Control, and ACM/IEEE International Conference on Cyber-Physical Systems, (HSCC/ICCPS 2026), Saint Malo, France, May 11-14 2026.

Nikhil Singh, Indranil Saha. “*Reward Generation for Model-Free Reinforcement Learning from Formal Specifications.*” To be published in the Journal of Artificial Intelligence Research (JAIR 2026).

Nikhil Singh, Indranil Saha. “*Frugal Actor-Critic: Sample Efficient Off-Policy Deep Reinforcement Learning Using Unique Experiences.*” International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024), Auckland, New Zealand, May 06-10, 2024. <https://dl.acm.org/doi/10.5555/3635637.3663037>.

Code at <https://github.com/iitkcpslab/FAC>

Nikhil Singh, Indranil Saha. “*STL-Based Synthesis of Feedback Controllers Using Reinforcement Learning.*” AAAI Conference on Artificial Intelligence (AAAI 2023), Washington D.C., February 07-14, 2023. <https://ojs.aaai.org/index.php/AAAI/article/view/26764>.

Code at <https://github.com/iitkcpslab/rlstl>

Nikhil Singh, Indranil Saha. “*Specification Guided Automated Synthesis of Feedback Controllers.*” ACM SIGBED International Conference on Embedded Software (EMSOFT 2021), ACM/IEEE, virtual conference, October 08-15, 2021.

Published in ACM Transactions on Embedded Computing Systems (2021) <https://dl.acm.org/doi/10.1145/3477011>.

Code at <https://github.com/iitkcpslab/SCoSyn>

Nikhil Singh, Indranil Saha. “*Specification Guided Automated Debugging of CPS Models.*” International Conference on Embedded Software (EMSOFT2020), ACM/IEEE, virtual conference, September 20-25, 2020.

Published in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (2020) <https://ieeexplore.ieee.org/document/9211574/>.

Code at <https://github.com/iitkcpslab/Blars>

* The author names appear in lexicographic order.

AWARDS & RECOGNITIONS

Fellowship for Academic and Research Excellence (Jan’25 - Aug’25).

Prime Minister’s Research Fellowship (Dec’20 - Dec-23).

ACM India IARCS travel grant (2024).

Invited to **Google Research Week 2023 & 2024**.

Nominated by the Department of CSE, IIT Kanpur for **Cadence Gold Medal** and **SIIC Student Innovation Award** for MS thesis.

First prize in Data structure & algorithm based event *CODATHON* in OJASS’13.

Third prize in Coding based event *CODEMANIA* in Ojass’ 13.

Second prize in the event *CODESTORM* conducted by the Computer Society of India at NIT-Jamshedpur held in March-2012.

First prize in the event *LET US C* conducted by the Computer Society of India at NIT-Jamshedpur held in April-2011.

TALKS

Frugal Actor-Critic: Sample Efficient Off-Policy Deep Reinforcement Learning Using Unique Experiences	<i>AAMAS-24, Auckland, New Zealand,</i>	May’24
Specification Guided Controller Synthesis Using RL	<i>CSE Dept. IIT Kanpur</i>	Oct’23

	Learning-based Controller Synthesis <i>Ericsson AI Research</i>	Mar'23
	STL-Based Synthesis of Feedback Controllers Using Reinforcement Learning <i>AAAI-23, Washington D.C.</i>	Feb'23
	Introduction to Controller Synthesis <i>CSE Dept. IIT Kanpur</i>	Oct'21
	Specification Guided Automated Synthesis of Feedback Controllers <i>EMSOFT-21, Online</i>	Oct'21
	Specification-Guided Automated Debugging of CPS Models <i>EMSOFT-20, Online</i>	Sep'20
TEACHING	Instructor Introduction to PyTorch (PMRF Lecture series) <i>Institute of Smart Structures and Systems (ISSS)</i>	Sep'23 - Dec'23
	Instructor Controller Synthesis using Python (PMRF Lecture series) <i>Institute of Smart Structures and Systems (ISSS)</i>	Mar'23 - Jul'23
	Instructor Introduction to Controller Synthesis (PMRF Lecture series) <i>Institute of Smart Structures and Systems (ISSS)</i>	Sep'22 - Dec'22
	Tutor Introduction to Computing (ESC101) <i>Indian Institute of Technology, Kanpur</i>	May'22 - Jun'22
	Teaching Assistant Software Development (CS253) <i>Indian Institute of Technology, Kanpur</i>	Jan'22 - May'22
	Teaching Assistant Autonomous Cyber-Physical Systems (CS659A) <i>Indian Institute of Technology, Kanpur</i>	Jan'21 - May'21
	Teaching Assistant Embedded and Cyber-Physical Systems (CS637A) <i>Indian Institute of Technology, Kanpur</i>	Sept'20 - Dec'20
	Teaching Assistant Formal Methods for Robotics and Automation (CS638A) <i>Indian Institute of Technology, Kanpur</i>	Jan'20 - May'20
PROFESSIONAL SERVICES	PC Member ECAI-2025, AAAI-2025	
	Reviewer ICC-2024	